

Supplementary Methods and Results

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A1 Data, time axes and experimental conditions

This appendix supports the eight-page course report. The frozen 31-page laboratory report retains the broader investigations and outstanding laboratory questions. Numerical results here come from the same saved studies, plus the September 13 trajectory-distribution follow-up. No new neural training was needed for that follow-up.

Table A1. Preprocessing stages. The onset-to-end crop used for features is distinct from the go-to-end window supplied to the trajectory models.

Stage	Input	Output and purpose
Loading	CSV frame counters, 3-D positions; paired MATLAB target trace	Trial objects with task metadata and event alignment
Regularization	Irregular/repeated frame entries	240 Hz grid after duplicate averaging and linear gap filling
Filtering	Regular 3-D positions	Fourth-order 10 Hz forward-backward Butterworth positions
Window	Target-motion onset and final tracker sample	Go-to-end path, including waiting after target motion
Resampling	Variable-length filtered path	100 phase samples, then translation to a zero start
Planar input	100 x 3 array	x-y array of 100 x 2; 200 time-interleaved values for neural models

The condition vector contains three one-hot starting-category entries, a side indicator and measured target speed divided by the 1920-pixel screen width. The nominal category settings were 120, 140 and 160 mm from screen center, associated with speed ranges 255-300, 298-350 and 340-400 mm/s in the laboratory stimulus description. The executed speed supplied to the encoder is derived from the target trace, not a one-hot speed class. Starting category and speed are coupled by design.

The appearance-to-motion foreperiod ranged from 0.20 to 0.50 s (median 0.35 s). The current modeled window begins at target motion, so it retains early movement already present at that point but excludes earlier appearance-to-motion kinematics. A three-frame threshold crossing in 3-D speed identifies finger onset. When no crossing is found, code falls back to target motion; a zero label can also occur when the threshold is crossed at the start. These cases must not be equated with noise or a behavioral strategy.

The recorded spatial coordinates are in cm and x-y is the task plane. Nominal starting settings are in mm relative to screen center. The two low-count participants had incomplete recording availability; their retained trials remain valid. Source: `src/data_loading.py`, `src/preprocessing.py`, `src/trajectory_view.py`; `production/cohort_trial_counts.csv`.

A2 Spline + PCA: the complete matrix pipeline

For one trial, x and y each have 100 samples. A fixed cubic B-spline basis with five interior knots has $5 + 3 + 1 = 9$ basis functions, giving B of shape 100×9 . Least squares solves $B \mathbf{c}$ approximately equal to x separately for each axis. The 18 coefficients are ordered as all nine x values followed by all nine y values. This is a different ordering from the time-interleaved neural input.

$$\mathbf{c}_x = \arg \min_{\mathbf{c}} \|\mathbf{x} - B\mathbf{c}\|_2^2, \quad \mathbf{c}_y = \arg \min_{\mathbf{c}} \|\mathbf{y} - B\mathbf{c}\|_2^2.$$

$$\mathbf{z} = V^T[(\mathbf{c} - \mathbf{m}) \odot \mathbf{d}], \quad \hat{\mathbf{c}} = \mathbf{m} + \mathbf{d} \odot (V\mathbf{z}).$$

Table A2. n is 3 or 8 in the matched report. The number of retained PCA components is not the number of knots.

Object	Shape	Meaning
B	100×9	One common spline design matrix
$\mathbf{c}$	18	Coefficients for one trial: nine x , nine y
C training	$N \text{ training} \times 18$	One row per training trial
V	$18 \times n$	PCA directions learned from training rows
Z training	$N \text{ training} \times n$	Compressed training codes
Decoded coefficients	18	Inverse scale/center applied after PCA inverse
Decoded path	100×2	B times the two reconstructed coefficient blocks

Raw-coefficient PCA centers the coefficients but does not standardize each coefficient. The standardized variant additionally divides each coefficient by its training standard deviation. The implementation evaluates both variants on validation trajectory reconstruction MSE. PCA maximizes retained variance in its coefficient representation; it does not mathematically guarantee minimum trajectory MSE across every competing model. The validation choice matters because coefficient distance and sampled-path distance differ.

At reconstruction time the query trajectory is available, so its spline coefficients may be fitted before projection into the already learned PCA axes. No PCA axes or scalers are fitted to query data. At generation time only context coefficients produce the center; the query path is unavailable. Applying the inverse PCA transform produces coefficients directly, without fitting a spline to a query trajectory.

Timing regression

The timing design has $n + 5$ inputs per trial: its PCA scores and task conditions. Training-standardized inputs feed a Ridge regression with intercept, predicting two training-standardized log durations. Alpha is selected from 0.01, 0.1, 1, 10 and 100 by validation MSE in standardized log-duration space. The fitted model remains based on the 17 training participants. Timing targets are not appended to the trajectory PCA coefficient matrix in the final evaluated configuration.

$$\|Y - UW - \mathbf{1}\mathbf{b}^T\|_F^2 + \alpha \|W\|_F^2.$$

Source: `src/confirmatory_spline.py` and the final strategy evaluation runner. In the displayed objective U is the standardized input matrix, Y the two-column target matrix, W the regression weights, and $\mathbf{b}$ the unpenalized intercept. Generation can therefore change spline durations with task conditions even though the spatial spline decoder ignores them.

A3 Neural objective and inference stages

All coordinate means and scales are learned on training trials. Each neural encoder receives 200 standardized coordinates and five condition slots. CVAE and CAE use the actual slots; VAE sets them to zero at both encoder and decoder. Two hidden layers of 256 ReLU units produce a deterministic code (CAE), or separate mean and log-variance heads (VAE/CVAE). Variational training draws standard-normal epsilon and reparameterizes the latent sample. The decoder has two 256-unit ReLU layers and coordinate and timing heads.

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, I).$$

$$\mathcal{L} = \|\mathbf{v} - \hat{\mathbf{v}}\|_2^2 + 20\|\mathbf{y} - \hat{\mathbf{y}}\|_2^2 + \beta D_{\text{KL}}[q(\mathbf{z} | \mathbf{v}, \mathbf{c}) \| \mathcal{N}(\mathbf{0}, I)].$$

$$D_{\text{KL}} = \frac{1}{2} \sum_{k=1}^n (\mu_k^2 + \sigma_k^2 - 1 - \log \sigma_k^2).$$

The reconstruction term sums 200 squared standardized coordinate errors. The timing term sums two squared standardized log-duration errors and multiplies by 20. KL sums over n latent dimensions. The batch then averages the per-trial objective. For CAE the KL term is absent. Beta starts at zero and reaches one at epoch 51. Validation uses the full beta = 1 objective, avoiding a different target simply because the training warm-up is incomplete.

Table A3. The word VAE does not imply that all evaluation stages sample from the standard-normal training prior.

Stage	Encoder sees	Latent value	Outputs and use
Training	Observed training shape; conditions where enabled	Posterior draw, or deterministic CAE code	Coordinate and timing losses plus KL where applicable
Reconstruction	Observed test shape; conditions where enabled	Posterior mean, or deterministic code	Path compared with this observed path
Context encoding	Context shapes only; their conditions where enabled	Mean code for every context trial	Average codes into participant center
Generation	No query path enters encoder	Draw around context center with training covariance	New path and predicted durations

Adam uses learning rate 0.001, batch size 64 and gradient clipping at norm 5. Weighted sampling gives each training participant equal expected contribution. Validation controls learning-rate reduction, checkpoint selection and early stopping with patience 25, for at most 150 epochs. The three random seeds quantify some training variation; they are averaged within a participant before model tests and are not counted as independent participants.

Durations are inverse-standardized and converted by $\exp(\log \text{target}) - 0.001$, with negative outputs clipped to zero. The generation feature pipeline additionally floors movement duration at 1 ms and limits the initiation fraction used for cropping to 0.98. No upper clipping removed extreme positive predictions from the main evaluation. The coordinate encoder never receives the two observed timing labels.

Source: `src/vae_model.py`, `src/train.py`, `src/confirmatory_evaluation.py` and `scripts/run_review_controls.py`. Condition masking is an implemented ablation; it does not establish that conditions have no relation to human movement.

A4 Context enrollment and stochastic generation

A test participant belongs to one held-out rotation. A fixed stratified split allocates approximately half of the participant's trials to context and half to query. Each model uses that same assignment. Averaging context posterior means produces θ_s ; spline and CAE use their deterministic codes. A participant can therefore be unseen during population training while still supplying recordings for enrollment.

$$\theta_s = \frac{1}{|C_s|} \sum_{i \in C_s} \mu_i, \quad \mathbf{z}^{(b)} \sim \mathcal{N}(\theta_s, \Sigma_{\text{train}}).$$

$$\Sigma_{\text{train}} = \text{Cov}_{\text{train}}(\mu_i - \bar{\mu}_{s(i)}) + \text{diag}(\mathbb{E}_{\text{train}}[\sigma_i^2]) + 10^{-6}I.$$

The first covariance term pools deviations of each training code from that training participant's average code. For variational models the diagonal of mean posterior variances is added; spline and CAE omit it. All models add $10^{-6}I$. The participant's context code center changes at enrollment, while this training covariance and the decoder remain fixed. The training sample covariance is not a separate variance estimated from each test participant's few context trials.

Each participant receives 120 generated trials. The generation procedure uses task conditions sampled from the query-condition mixture, allowing a matched-condition evaluation. It withholds query positions and durations. VAE ignores the sampled task conditions, spline uses them for timing, and CVAE/CAE use them in decoding. This is not a test of extrapolation to unseen conditions. Three neural training seeds are averaged; spline uses its single frozen fitting and sampling realization.

The trajectory follow-up reproduced the previous feature scores from the generated arrays to machine precision before evaluating new endpoints. It reused 72 frozen neural checkpoints for $n = 3/8$ and recreated the classical fits. Condition Ridge required the historical two-thread numerical setting to reproduce the near-singular fit. No neural training changed, and the laboratory report remained frozen. Saved arrays under `production/assets/trajectory_distribution_2026_09_13` allow the spatial analysis to be repeated without retraining when the local cache is available.

Personal-center replacement control

The own-context center is compared with the training-population center and the average performance of six other test-participant centers from the same rotation. Decoder, covariance, random latent noise and task draws remain fixed. This tests whether changing the center to the correct participant improves this generator. It does not test long-term stability or identify physical meanings for the individual latent axes. The control was evaluated on feature distances for CVAE $n = 3/8$ and VAE $n = 8$, not for every model/capacity and not for the new full-trajectory endpoints.

Sources: `src/context_query.py`, `scripts/run_review_controls.py` and `production/trajectory_distribution_2026_09_13/protocol.json`. Generation inherits the earlier character-sum participant seed convention, which can give different identifiers the same random-number stream. The saved protocol discloses this reproducibility limitation. It does not expose query trajectories to model fitting.

A5 The eleven feature distributions

Table A4. The count is a speed-peak heuristic, not the number of fitted minimum-jerk components.

Feature	Definition / role
Initiation time	Seconds from target motion to detected finger onset
Movement time	Seconds from detected finger onset to final sample
Peak speed	Largest planar speed after feature-stage processing
Relative time to peak speed	Peak-speed time expressed as fraction of movement duration
Path length	Sum of planar segment lengths
Straight-line distance	Distance between movement-segment start and end
Curvature index	Path length divided by straight-line distance
Maximum lateral deviation	Largest deviation from the start-to-end straight line
Speed-peak count	Heuristic peaks with prominence 10% of maximum speed and about 50 ms separation; minimum reported count one
Endpoint x	Final x coordinate
Endpoint y	Final y coordinate

Recorded features use the observed onset-to-end segment and its measured duration. Generated features crop the go-to-end trajectory using predicted initiation/(initiation + movement time), capped at 0.98. Both feature representations contain 100 movement-phase points. Their physical sampling rate is 99 divided by movement duration. The feature-stage low-pass cutoff is $\min(10 \text{ Hz}, 0.45 \text{ times that sampling rate})$, and numerical gradients provide velocities. Matching this downstream processing was essential for fair recorded/generated comparisons.

$$D_{sf} = \sup_x |F_{sf}(x) - G_{sf}(x)|, \quad \overline{D}_s = \frac{1}{11} \sum_{f=1}^{11} D_{sf}.$$

An empirical cumulative distribution $F(x)$ is the fraction of observations at or below x . For one feature and participant, KS is the largest vertical separation between the recorded and generated cumulative curves. We report its statistic, which ranges from zero to one. We then average eleven statistics within a participant. This is not an average of eleven p-values, and the BH family does not contain eleven separate KS tests per participant.

Energy and MMD instead receive the entire eleven-value feature vector per trial. Training means and standard deviations put these quantities on comparable numerical scales, with the peak-count scale floored at one. Correlation still matters: redundant coordinates can emphasize the same underlying property in a Euclidean distance. Appendix A11 applies a training-only redundancy rule and repeats the evaluation with nine features. This changes the multivariate endpoint. It should not be chosen merely to obtain smaller corrected p-values.

Sources: `src/features.py`, `src/confirmatory_evaluation.py` and corrected feature-processing paths in `scripts/run_review_controls.py`. Timing errors and spatial errors can both change the generated features, whereas the full go-to-end path distances in the new analysis do not use predicted durations to crop trajectories.

A6 Distances and inferential statistics

$$E = 2 \overline{\|X - Y\|} - \overline{\|X - X'\|} - \overline{\|Y - Y'\|}.$$

$$\widehat{\text{MMD}}_u^2 = \frac{\sum_{i \neq j} k(X_i, X_j)}{m(m-1)} + \frac{\sum_{i \neq j} k(Y_i, Y_j)}{r(r-1)} - \frac{2 \sum_{i,j} k(X_i, Y_j)}{mr}.$$

For recorded vectors X and generated vectors Y , energy uses twice the mean cross Euclidean distance minus the two mean within-sample distances. All ordered within pairs, including diagonal zeros, enter this empirical V-statistic. There is no extra square root of the resulting statistic. Subtracting within-sample distances matters: a collapsed mean path can have small distance to many observations yet fail to reproduce their variation.

MMD uses $k(x,y) = \exp(-\gamma \text{ squared Euclidean distance})$. Each within-sample term excludes the diagonal and divides by the number of ordered distinct pairs, whereas the cross term includes every pair. This unbiased squared estimator can be slightly negative at finite sample sizes; it is not a negative physical distance. Gamma is the inverse median positive squared distance among a deterministic sample of at most 512 training trials. Feature and path analyses use their own training representations and fixed bandwidths, shared by all models within a rotation.

Raw path vectors contain the flattened ordered 100×2 coordinates divided by $\sqrt{200}$. Each Euclidean pair distance therefore equals RMS coordinate difference in cm. The axis-balanced sensitivity additionally divides x and y by their respective training RMS deviations around the training mean path. It does not standardize every time point separately or align paths by dynamic time warping. Feature geometry instead uses its eleven training-standardized features. Absolute values across these representations should not be directly compared.

The unit of inference

For a model pair we have 28 paired participant scores after averaging seeds within participant. The Pratt Wilcoxon procedure ranks absolute paired differences, retaining zeros in ranking but excluding their signed contribution. A two-sided test evaluates a signed-rank null; its usual location interpretation relies on symmetry of the difference distribution. This is not a normality test or a test on thousands of supposedly independent trials. Rounding differences to 12 decimals prevents numerical round-off alone from breaking ties.

BH sorts p-values and controls the false-discovery rate under its applicable dependence assumptions. Holm provides a more conservative familywise-error sensitivity. Neither correction makes overlapping trained models or cohort-based model selection independent. We report exploratory evidence and both corrections rather than interpreting a threshold as proof. Nonsignificance is not equivalence. The seven original endpoints were trajectory MSE, initiation MAE, movement MAE, mean KS, speed-peak-count KS, feature energy and feature MMD squared.

Table A5. Separate analysis families. The new 80-test correction includes both geometries together. No new $n=3$ -versus- $n=8$ or empirical-reference significance test is claimed.

Family	Number	Definition
Original model comparisons	140	10 model pairs x 2 dimensions x 7 endpoints
New path comparisons	80	10 pairs x 2 dimensions x 2 metrics x 2 geometries
CVAE center controls	12	2 dimensions x 2 replacement controls x 3 feature metrics
VAE center controls	6	$n=8$ x 2 replacement controls x 3 feature metrics

A7 Selected model comparisons and exact corrected values

Every row compares VAE with the named model at the same dimension. Direction must be read with the reported means in the main paper. q denotes BH over the full original family of 140 tests; Holm p uses that same family. Display rounding does not determine significance.

Table A6. All 24 VAE comparisons against the other three compact models on the four main original endpoints.

n	Comparator	Endpoint	BH q	Holm p
3	Spline + PCA	Reconstruction MSE	7.56e-05	0.00248
3	Spline + PCA	Mean feature KS	2.53e-05	0.000781
3	Spline + PCA	Feature energy	0.000105	0.00362
3	Spline + PCA	Feature MMD squared	2.8e-05	0.000902
3	CAE	Reconstruction MSE	0.0369	1
3	CAE	Mean feature KS	0.783	1
3	CAE	Feature energy	0.035	1
3	CAE	Feature MMD squared	0.0233	0.804
3	CVAE	Reconstruction MSE	0.00163	0.0578
3	CVAE	Mean feature KS	0.0242	0.832
3	CVAE	Feature energy	0.991	1
3	CVAE	Feature MMD squared	0.944	1
8	Spline + PCA	Reconstruction MSE	0.000476	0.0168
8	Spline + PCA	Mean feature KS	2.25e-05	0.000674
8	Spline + PCA	Feature energy	2.96e-06	7.21e-05
8	Spline + PCA	Feature MMD squared	5.34e-06	0.000136
8	CAE	Reconstruction MSE	4.97e-08	1.04e-06
8	CAE	Mean feature KS	0.000168	0.00591
8	CAE	Feature energy	0.000275	0.00958
8	CAE	Feature MMD squared	8.4e-05	0.00283
8	CVAE	Reconstruction MSE	0.94	1
8	CVAE	Mean feature KS	0.137	1
8	CVAE	Feature energy	0.0257	0.875
8	CVAE	Feature MMD squared	0.0518	1

The condition-only Ridge reference had MSE 1.233795, mean feature KS 0.277264, feature energy 0.818722 and feature MMD squared 0.125854. Its full-path raw energy was 0.252014 and MMD squared 0.109010. It has no latent dimension and is reused in both dimension-specific comparison families. Complete all-pairs values remain in `audit_2026_09_08/all_pairs_140.csv` and `trajectory_distribution_2026_09_13/paired_comparisons.csv`.

A8 Full-path robustness and empirical references

Table A7. Axis-balanced sensitivity using training-only scales. Lower is better.

Model	n	Axis-balanced energy	Axis-balanced MMD squared
Spline + PCA	3	0.1230	0.0730
CAE	3	0.1277	0.0786
CVAE	3	0.1109	0.0610
VAE	3	0.1092	0.0608
Spline + PCA	8	0.0912	0.0545
CAE	8	0.0866	0.0531
CVAE	8	0.0594	0.0270
VAE	8	0.0569	0.0243

Table A8. Variational models versus spline, corrected over all 80 path tests. At n = 8 every listed advantage survives Holm. The n = 3 conclusion changes with metric/scaling.

n	Pair	Metric	Geometry	BH q	Holm p
3	CVAE / spline	energy	Raw	0.205	1
3	CVAE / spline	mmd2	Raw	0.00184	0.0363
3	CVAE / spline	energy	Balanced	0.387	1
3	CVAE / spline	mmd2	Balanced	0.148	1
3	VAE / spline	energy	Raw	0.0647	1
3	VAE / spline	mmd2	Raw	0.00184	0.0363
3	VAE / spline	energy	Balanced	0.219	1
3	VAE / spline	mmd2	Balanced	0.0622	1
8	CVAE / spline	energy	Raw	1.19e-06	7.72e-06
8	CVAE / spline	mmd2	Raw	1.26e-06	1.03e-05
8	CVAE / spline	energy	Balanced	0.00012	0.00259
8	CVAE / spline	mmd2	Balanced	5.29e-05	0.00112
8	VAE / spline	energy	Raw	1.97e-06	2.21e-05
8	VAE / spline	mmd2	Raw	8.34e-07	4.07e-06
8	VAE / spline	energy	Balanced	1.36e-05	0.000253
8	VAE / spline	mmd2	Balanced	5.61e-06	9.25e-05

Stored-context resampling selected 120 paths with replacement for each participant and averaged three stable SHA-derived random seeds. In raw geometry it achieved energy 0.032847 and MMD squared 0.005407. Repeating the context mean gave energy 0.462058 and MMD squared 0.245241. Neither reference compresses a participant into n numbers, and neither comparison received a significance test. The mean-only reference is a collapse diagnostic, not a candidate stochastic model.

The independent verifier checked 184 decoded-array cases and all 80 paired tests. Existing feature scores were reproduced to a largest absolute difference of 4.44×10^{-16} . These computational checks establish consistency with the saved analysis, not independent scientific replication. Protocol, complete scores and verification records accompany the results folder.

A9 Dashboard and reproducibility

The dashboard defaults to VAE $n = 8$, the lowest mean feature-discrepancy configuration in the matched comparison. CVAE and spline offer $n = 2, 3, 4, 8$, while CAE and VAE offer $n = 3, 8$. These are the evaluated model choices; no $n = 16$ result is exposed. Models have different conditioning behavior, so moving a condition control is not a validated causal intervention on human behavior.

The Generate section shows live examples from fold 0, seed 42. Validation distributions use the available exported seed-42 examples across four rotations, while score cards summarize the study seeds. A demonstration should distinguish these illustrative trajectories from cohort-level significance. The report's subject01 figure uses fold 2 and therefore need not match the live fold-0 examples.

PowerShell launch

From the repository folder, run:

```
python -m streamlit run src/confirmatory_dashboard.py --server.port 8510 --server.address 127.0.0.1 --browser.gatherUsageStats false
```

Open <http://127.0.0.1:8510> in a browser. Keep the terminal open during the demonstration and press Ctrl+C there to stop the server. The standalone archive in output/release includes its own launch instructions when a full checkout and local training caches are unavailable.

Suggested short demonstration

Show VAE $n = 8$ and explain that the displayed paths are samples around a context-derived center. Change to $n = 3$ to illustrate the compact setting, without claiming this single example proves the cohort result. Show CVAE as the conditional alternative and spline + PCA as the statistical comparator. Finish on the benchmark view to return from individual trajectories to the study-level evidence.

Table A9. Navigation for checking methods against implementation and stored results.

Evidence or implementation	Location relative to repository
Main means and original tests	production/audit_2026_09_08/all_model_means.csv; all_pairs_140.csv
New trajectory analysis	production/trajectory_distribution_2026_09_13/
New analysis implementation	scripts/analyze_trajectory_distributions.py; src/trajectory_distribution.py
Generation and original evaluation	src/context_query.py; src/confirmatory_evaluation.py; scripts/run_review_controls.py
Model definitions	src/vae_model.py; src/baseline_spline.py; src/confirmatory_spline.py
Dashboard	src/confirmatory_dashboard.py; src/dashboard_models.py
Frozen extended report	production/Interception_Movements_Methods_Reconstruction_Review.pdf

A10 Secondary investigations and remaining interpretation

Timing and behavioral-summary probes

Timing heads typically had mean absolute errors of tens of milliseconds, while occasional large errors affected variance-explained measures. For VAE $n = 8$, initiation MAE was 35.98 ms and movement MAE 42.21 ms; the corresponding spline values were 55.58 and 50.21 ms. These timing reconstructions receive the observed shape and therefore are not prospective timing forecasts before a movement begins. Common-head comparisons in the laboratory report assess how changing the timing head changes representation comparisons.

Linear probes attempted to predict fourteen participant summaries: means and standard deviations of seven behavioral features. Directly computing those summaries on context trials yielded lower MAE on every target than each of the eight model/dimension probe configurations. In the pooled summaries, nine of fourteen VAE $n = 8$ targets and ten of fourteen CVAE $n = 3$ targets had negative R-squared. A useful center for a nonlinear generator therefore does not establish reliable linear access to every individual behavioral statistic. See `production/all_models_direct_context_summary.csv` and the frozen report, pages 18-22.

Conditioning and movement decomposition

Code checks verified that CVAE and CAE receive conditions at the intended inputs. Shuffling decoder conditions worsened mean reconstruction, so the routes were active. Condition-stratified feature tests did not establish corrected fidelity differences. These results neither prove a coding failure nor prove that experimental conditions are irrelevant. A larger causal conclusion would require a different evaluation design.

Minimum-jerk analyses fitted overlapping smooth velocity pulses under a matched downstream procedure. Their component-count total-variation discrepancies were approximately 0.214 (CVAE $n = 3$), 0.232 (CVAE $n = 8$) and 0.236 (VAE $n = 8$). None of the nine matched pairwise comparisons was significant after correction. Counts depended on fit assumptions and remained distinct from the simpler speed-peak feature used in the main report. The broader decomposition, event audit, clustering and enrollment experiments remain in the frozen report, pages 18-28, for laboratory discussion.

Scope for the next study

Promising follow-ups include richer participant-specific latent distributions, prospective evaluation after model selection, and testing across sessions or new task conditions. No current learning-curve experiment establishes that sample size caused the failures. Retaining a failed probe or a strong empirical reference helps identify what the compact model loses, rather than weakening the scientific value of reporting the actual results.

The course report selects the experiments that establish the modeling task, compare deep and statistical methods, evaluate generation and justify the dashboard. The laboratory report preserves the extended evidence while Jason reviews the experimental assumptions. Neither document substitutes for a new validation cohort.

A11 Training-only feature-redundancy sensitivity

A fixed protocol screened feature overlap using only the 17 training participants in each rotation. We calculated pooled Spearman correlations and the median signed correlation calculated separately within participants. A feature was removed only when an already retained feature had absolute correlation at least 0.90 on both measures, with matching signs. This rule required strong overlap both across training trials and within a typical participant. It does not establish that the retained features are independent.

The priority order favored initiation time, movement time, peak speed, relative time to peak, path length, curvature index, lateral deviation and speed-peak count, followed by endpoint x, straight-line distance and endpoint y. This order and threshold were recorded before computing the sensitivity results. Every rotation retained the same nine features, dropping straight-line distance and endpoint y in favor of path length.

Table A10. Training correlations with path length across four rotations. All values are positive. High correlation reflects overlap in this dataset, rather than a universal equivalence of these quantities.

Removed feature	Pooled rho range	Median within-person rho
Straight-line distance	0.933-0.971	0.953-0.978
Endpoint y	0.934-0.969	0.956-0.982

The frozen models generated the same 120 movements per test participant and seed as before. No neural model was retrained. The original eleven-feature scores reproduced to a maximum absolute difference of 4.44×10^{-16} . For the nine-feature view, training data defined the restricted feature scales and a new training-derived kernel bandwidth. We recomputed mean KS, energy and MMD for both views, averaging neural seeds within participant.

Table A11. Nine-feature discrepancies. Lower is better within this representation. Values cannot be used to claim absolute improvement over eleven features because removing coordinates changes the endpoint.

Model	n	Mean KS	Energy	MMD squared
Spline + PCA	3	0.3264	0.8824	0.1774
CAE	3	0.2814	0.7044	0.1412
CVAE	3	0.2726	0.6611	0.1293
VAE	3	0.2768	0.6504	0.1261
Spline + PCA	8	0.2551	0.6187	0.1204
CAE	8	0.2575	0.5377	0.1143
CVAE	8	0.2194	0.3913	0.0687
VAE	8	0.2156	0.3762	0.0662

The complete sensitivity family contained 72 tests: six compact-model pairs x two capacities x three metrics x two feature views. BH and Holm each corrected all 72 together. Both VAE and CVAE beat spline on every reduced-feature metric at both capacities under both corrections (largest Holm $p = 0.00332$). At $n = 8$, VAE versus CVAE had reduced-feature BH $q = 0.149$ for KS, 0.289 for energy and 0.428 for MMD. None was significant. The variational advantage over spline persisted; a unique VAE advantage did not.

The primary eleven-feature results and their historical 140-test correction remain unchanged. The separate sensitivity family must not be substituted for that correction. Removing correlated coordinates does not automatically reduce the number of model comparisons, and neither analysis consists of eleven feature-level KS p -values. Source: production/feature_redundancy_2026_09_14/; scripts/analyze_feature_redundancy.py.

A12 Reconstruction in spatial and phase coordinates

An x-y overlay displays spatial shape but can conceal differences in progression along the same path. The coordinate-versus-phase panels show the same recorded trial and the spline and VAE reconstructions at both capacities. Reconstruction MSE compares coordinates at corresponding phases, so these differences also contribute to its value.

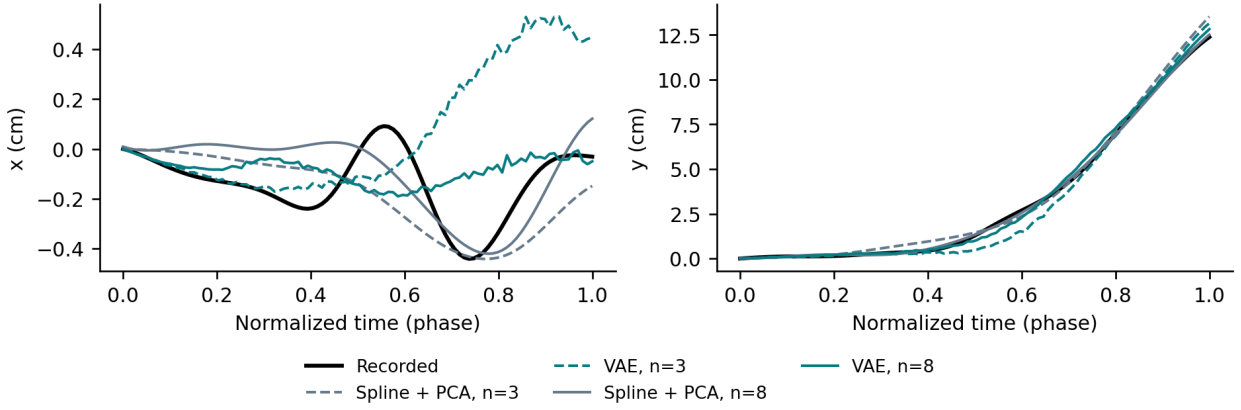


Figure A1. Subject01 median-duration trial, selected before inspecting errors. Black: recorded coordinates. Color identifies the model; dashed lines indicate $n = 3$ and solid lines $n = 8$. Phase is relative elapsed time in the modeled interval. Both panels retain coordinate units. The illustrative neural fit uses fold 2, seed 42.

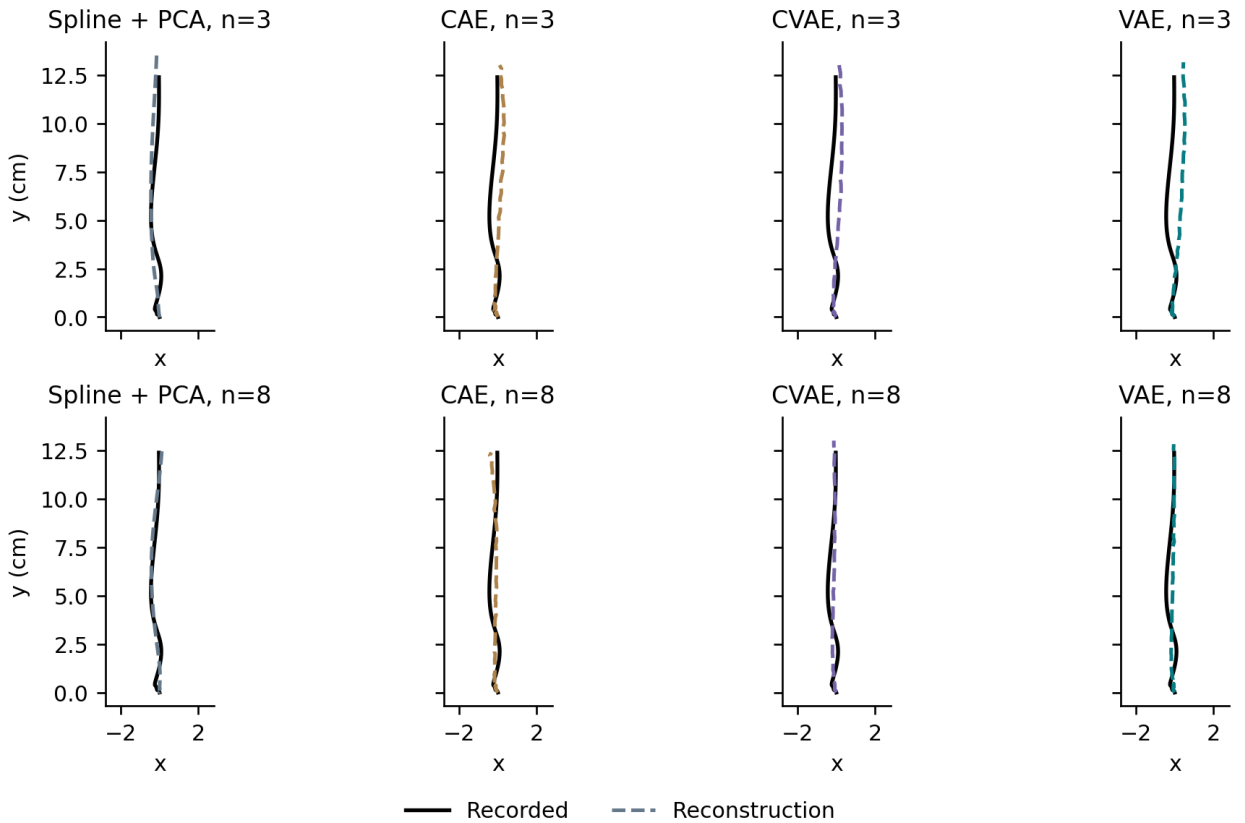


Figure A2. The same trial in x-y coordinates for all four models. Black: recorded; dashed: reconstruction. Equal x/y scale and shared limits retain physical proportions. An apparently close spatial overlay can still differ in the phase at which a position is reached.

A13 Recorded and generated feature distributions: n = 3

These empirical cumulative curves show all eleven features for the same identifier-selected participant. The recorded query set contains 90 trials and each model generates 120. Each feature curve is a marginal distribution; the joint energy and MMD scores also use dependence across features. The panels illustrate model behavior and do not replace the cohort-level paired tests.

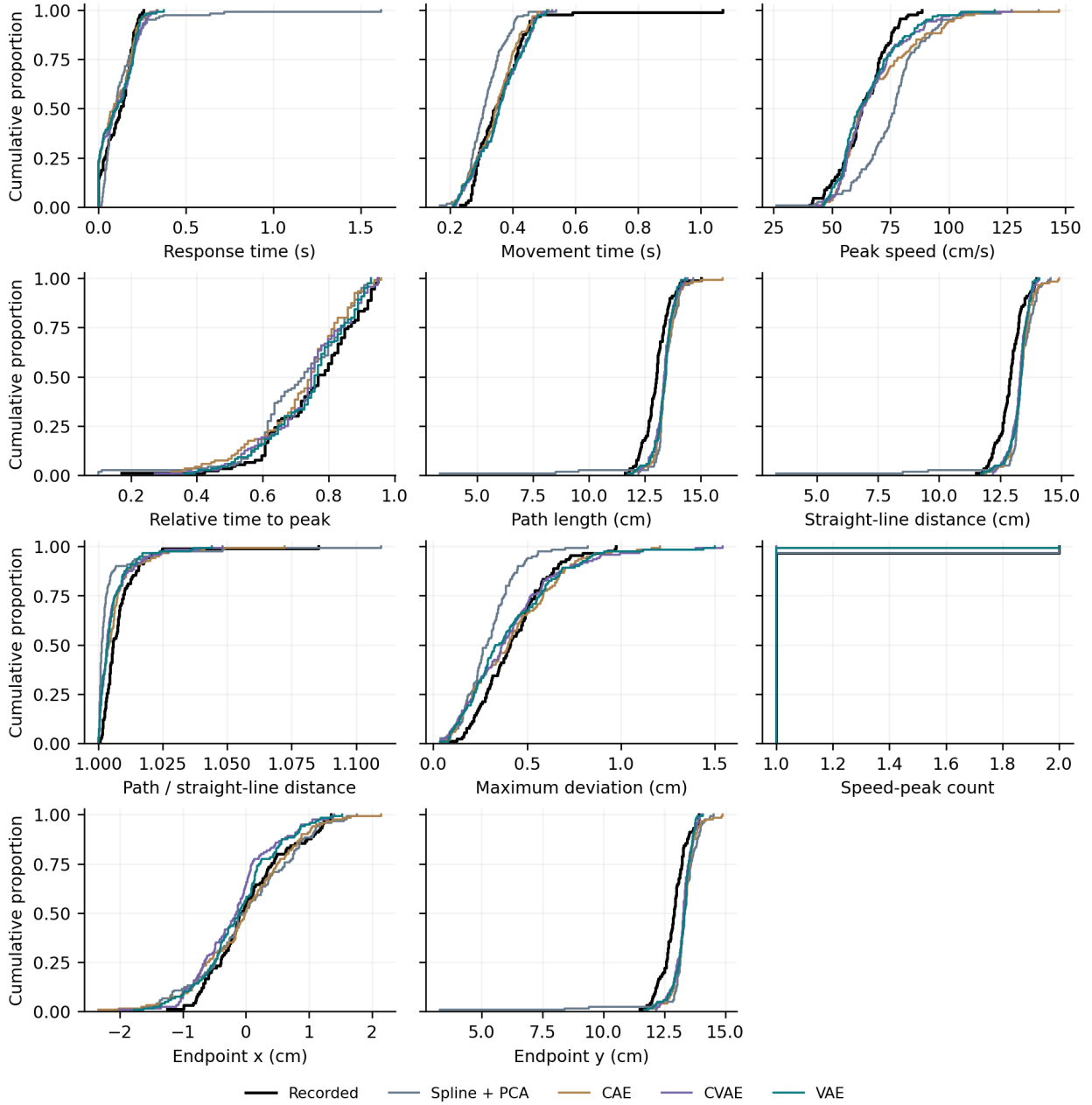


Figure A3. Subject01, n = 3, fold 2, neural seed 42. The fraction of values at or below each horizontal-axis value is shown vertically. Smaller maximum vertical separation from the black recorded curve means smaller KS for that feature and participant. All plotted values are included; feature axes have their own physical scales.

Straight-line distance and endpoint y are retained here to document the original eleven-feature analysis. The training-only reduction in A11 omits them from a separate sensitivity endpoint. Their removal was based on training correlation and a recorded feature priority, not on which query curves looked least favorable.

A13 Recorded and generated feature distributions: n = 8

These empirical cumulative curves show all eleven features for the same identifier-selected participant. The recorded query set contains 90 trials and each model generates 120. Each feature curve is a marginal distribution; the joint energy and MMD scores also use dependence across features. The panels illustrate model behavior and do not replace the cohort-level paired tests.

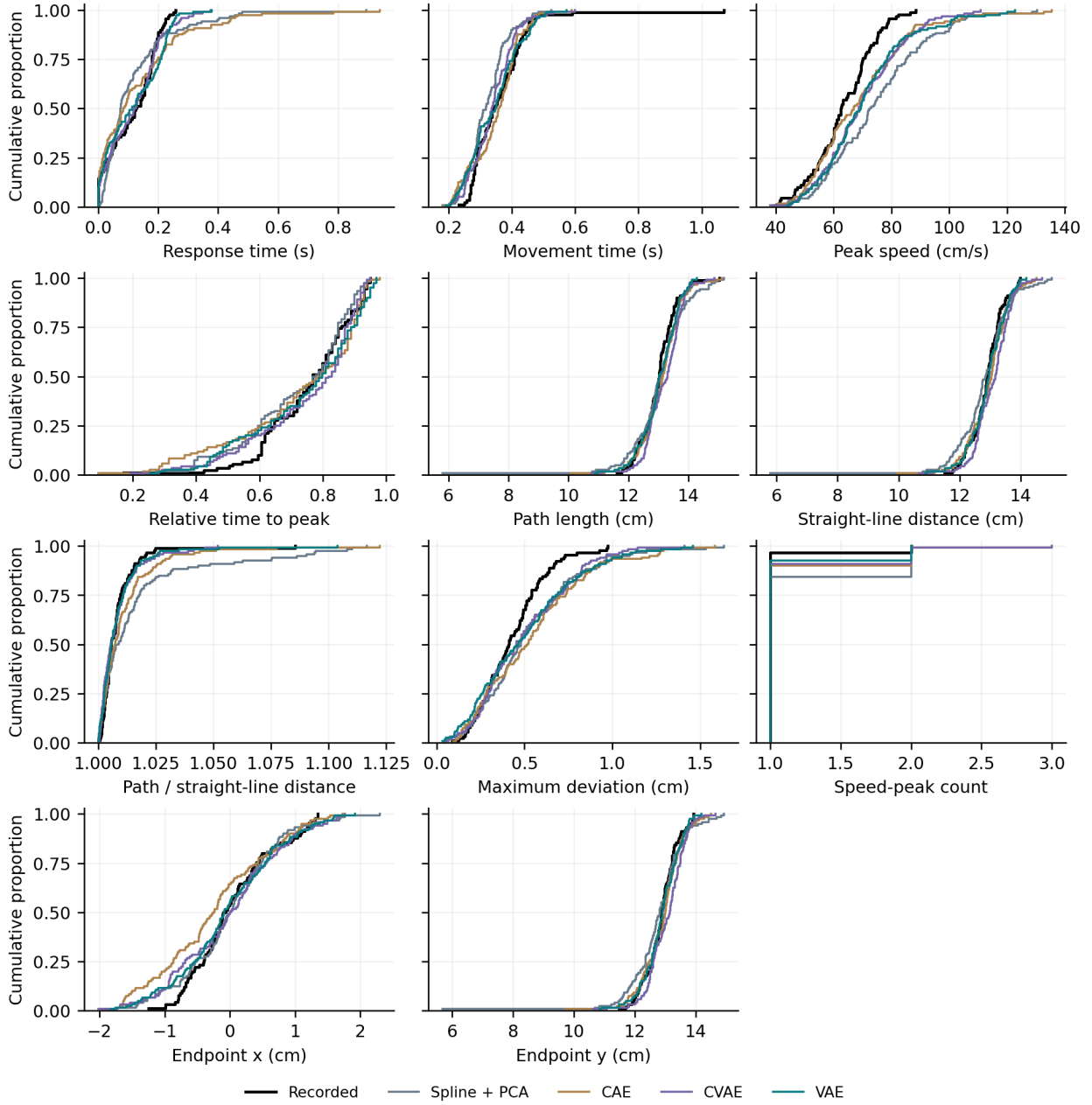


Figure A4. Subject01, n = 8, fold 2, neural seed 42. The fraction of values at or below each horizontal-axis value is shown vertically. Smaller maximum vertical separation from the black recorded curve means smaller KS for that feature and participant. All plotted values are included; feature axes have their own physical scales.

Straight-line distance and endpoint y are retained here to document the original eleven-feature analysis. The training-only reduction in A11 omits them from a separate sensitivity endpoint. Their removal was based on training correlation and a recorded feature priority, not on which query curves looked least favorable.